

Survey on Roman {2}-Domination

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Abstract: The notion of Roman {2}-domination was introduced in 2016 as a variant of Roman domination, a concept inspired by a defending strategy used by the emperor Constantine (272–337 AD) to protect the Roman Empire. Since then, a considerable number of papers on Roman {2}-domination and its variants have been published. In this paper, we survey published results on Roman {2}-domination as well as the main findings on Roman {2}-domination variants found in the literature. A list of open problems related to this notion and its variants are also given.

Keywords: Roman {2}-domination; Italian domination; Roman domination; domination

MSC: 05C69; 05C85; 05C05

1. Introduction

The concept of Roman {2}-domination in graphs was introduced in 2016 by Chellali in [1] as a variant of Roman domination. The original concept of Roman domination was inspired by the defensive strategies used by Emperor Constantine (272–337 AD) to protect the Roman Empire. The strategy was that (i) every city in the empire had to have a maximum of two legions stationed there. Moreover, (ii) every city without a legion had to be near another city with two armies, so in the event that an attack occurred against a city without an army, the city with two armies could dispatch one of its armies to defend the city. This strategy was then abstracted and studied as a mathematical concept in graph theory [2–6], and the aim was to develop such a strategy at minimum cost, that is, a strategy with the minimum number of armies.

In the literature, for example [7–14], Roman {2}-domination is also referred as Italian domination. The study of Roman {2}-domination and other variants of Roman domination has provided valuable insights into the properties of graphs and has found applications in various fields, including computer science, operations research, and network design.

Since 2016, many papers have been published on Roman {2}-domination, and the number is still rising quickly. Thus, it could be helpful for the research community to offer an overview of the findings and unresolved issues.

The objective of this project is to assemble nearly every published work on Roman {2}-domination. We think our list is pretty comprehensive after using different keywords to conduct an exploratory search in different databases. Our goal is also to limit the number of pages in the survey, so we only included the results that we thought were particularly interesting. There are still some issues and questions that remain unanswered even though many of the ones from the early studies have already been resolved. Additionally, the recently published papers present fresh viewpoints that raise new research challenges and open questions.

This article outlines some of the unresolved issues and concerns, notably those that the original paper's authors have drawn attention to. This means the selection is inevitably limited. Furthermore, we do not assert that the problems identified are the only significant unsolved problems.



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This paper is structured as follows: Section 2 is devoted to essential terms, definitions and symbols. Results, including algorithm complexity, on Roman $\{2\}$ -domination that have been published are shown in Section 3. Section 4 deals with variants of Roman $\{2\}$ -domination. We conclude this paper in Section 5.

2. Essential Definitions and Notations

In this section, necessary notations and definitions are provided. Unless stated otherwise, the graph $G = (V, E)$ is a finite, simple, and undirected graph. The order n of a graph G is $|V(G)|$. Let $u, v \in V(G)$; we say that u is *adjacent* to v , or u and v are *neighbors* if $uv \in E(G)$. The *open neighborhood* $N(v)$ of a vertex v is the set of all neighbors of v , while the *closed neighborhood*, $N[v]$, of a vertex v is $N(v) \cup \{v\}$. The *degree* $d(v)$ of a vertex v is $|N(v)|$. The *maximum degree* $\Delta(G)$ of a graph G is $\max\{d(v) | v \in V(G)\}$, and the *minimum degree* $\delta(G)$ of G is $\min\{d(v) | v \in V(G)\}$. A path of order n is denoted by P_n , while a cycle of order n is denoted by C_n . A subgraph H of G is called an *induced subgraph* of G , if for every pair of vertices $u, v \in V(H)$, u and v are adjacent in H if and only if they are adjacent in G . A vertex $v \in V(G)$ is called an *isolated vertex* if $d(v) = 0$.

A subset $S \subseteq V(G)$ is called a *dominating set* of G if every vertex $v \in V(G) \setminus S$ is adjacent to a vertex in S . The *domination number* of G , denoted by $\gamma(G)$, is the minimum cardinality of a dominating set of G . Let A be a finite subset of $\mathbb{Z}$. The *weight*, $w(f)$, of a function $f : V(G) \rightarrow A$ is the sum $\sum_{v \in V(G)} f(v)$. Every such function f corresponds to the partition $(V_i^f | i \in A)$, where $V_i^f = \{v \in V(G) | f(v) = i\}$. A function $f : V(G) \rightarrow \{0, 1, 2\}$ is called a *Roman dominating function* on G if every $v \in V_0^f$ is adjacent to a vertex $u \in V_2^f$. The *Roman domination number* of G , denoted by $\gamma_R(G)$, is the minimum weight of a Roman dominating function on G . Roman domination was originally a defending strategy used by Emperor Constantine (272–337 AD) to protect the Roman Empire, then it was studied as a mathematical concept in [2–6].

A vertex $v \in V(G)$ is said to be *undefended* with respect to a function $f : V(G) \rightarrow \{0, 1, 2\}$ if $v \in V_0^f$ and $N(v) \cap (V_1^f \cup V_2^f) = \emptyset$. A function $f : V(G) \rightarrow \{0, 1, 2\}$ is called a *weak Roman dominating function* on G if for every $v \in V_0^f$ there exists $u \in N(v) \cap (V_1^f \cup V_2^f)$ such that the function $g : V(G) \rightarrow \{0, 1, 2\}$ defined by $g(v) = 1$, $g(u) = f(u) - 1$, and $g(w) = f(w)$ for every $w \in V(G) \setminus \{v, u\}$ has no undefended vertex. The *weak Roman domination number* of G , denoted by $\gamma_r(G)$, is the minimum weight of a weak Roman dominating function on G . Weak Roman domination was introduced in [15] to reduce the cost of defending the Roman Empire. To read more on weak Roman domination, we refer the reader to [16].

Let $P(\{1, 2\})$ be the set of all subsets of $\{1, 2\}$. A function $f : V(G) \rightarrow P(\{1, 2\})$ is called a *2-rainbow dominating function* on G if for every $v \in V(G)$ with $f(v) = \emptyset$, $\cup_{u \in N(v)} f(u) = \{1, 2\}$. The *weight*, $w(f)$, of a 2-rainbow dominating function f is $\sum_{v \in V(G)} |f(v)|$. The *2-rainbow domination number* of G , denoted by $\gamma_{r2}(G)$, is the minimum weight of a 2-rainbow dominating function on G . The 2-rainbow domination was introduced in [17]. To read more on 2-rainbow domination, we refer the reader to [18–20].

Roman $\{2\}$ -domination (also called Italian domination) was introduced in [1] as a variant of Roman domination. A function $f : V(G) \rightarrow \{0, 1, 2\}$ on G is called a *Roman $\{2\}$ -dominating function* if every $v \in V_0^f$ is adjacent to a vertex in V_2^f or is adjacent to at least two vertices in V_1^f . The *Roman $\{2\}$ -domination number* of G , denoted by $\gamma_{R2}(G)$, is the minimum weight of a Roman $\{2\}$ -dominating function on G . In some articles, the Roman $\{2\}$ -domination number is called the Italian domination number, and they denote it by $\gamma_I(G)$.

The Roman $\{2\}$ -domination number was independently introduced in [17] under the name “weak $\{2\}$ -domination number”; it was introduced as a tool to provide a linear algorithm to find a 2-rainbow dominating set of minimum size for trees.

3. Roman $\{2\}$ -Domination

This section is devoted to Roman $\{2\}$ -domination results found in the literature, and it is divided to three subsections. The first subsection is for the general results, the second one is for the results related to special classes of graphs, and the last subsection is for the algorithmic complexity of Roman $\{2\}$ -domination problems.

3.1. General Results

We start this subsection by providing the main results obtained in the first paper on Roman $\{2\}$ -domination.

Proposition 1 ([1]). *If G is a connected graph of order n , then $\gamma_{R2}(G) \geq \frac{2n}{\Delta(G)+2}$.*

Theorem 1 ([1]). *Let G be a graph. Then*

$$\gamma(G) \leq \gamma_r(G) \leq \gamma_{R2}(G) \leq \gamma_{r2}(G) \leq \gamma_R(G) \leq 2\gamma(G).$$

The 2-rainbow domination number is a sharp upper bound for the Roman 2-domination number as the following theorem indicates:

Theorem 2 ([1]). *If G is a tree or a cactus graph with no even cycle, then $\gamma_{R2}(G) = \gamma_{r2}(G)$.*

The weak Roman domination number is a sharp lower bound as $\gamma_r(G) = \gamma_{R2}(G)$ when G is a star graph of order $n \geq 3$ [1].

Maryam Hajibaba and Nader Rad [11] characterized nontrivial connected graphs G satisfying $\gamma(G) = \gamma_{R2}(G)$.

Problem 1. *Characterize graphs G satisfying $\gamma_{R2}(G) = \gamma_{r2}(G)$.*

Problem 2. *Characterize graphs G satisfying $\gamma_{R2}(G) = \gamma_r(G)$.*

An upper bound and a lower bound of the Roman $\{2\}$ -domination number in terms of a Roman domination variant called double Roman domination number, $\gamma_{dR}(G)$, are provided in [8]. Double Roman domination number was introduced in [21].

Theorem 3 ([8]). *For every graph G , $\frac{\gamma_{dR}(G)}{2} \leq \gamma_{R2}(G) \leq \frac{2\gamma_{dR}(G)}{3}$.*

The authors provided a constructive characterization of trees achieving the upper bound of Theorem 3. They also raised the following open problem:

Problem 3. *Characterize all trees achieving equality in the lower bound of Theorem 3.*

Sharp upper bounds of $\gamma_{R2}(G)$ in terms of the order of the graph were also studied in the literature.

Theorem 4 ([22]). *If G is a connected graph of order $n \geq 3$, then $\gamma_{R2}(G) \leq \frac{3}{4}n$.*

The authors characterized graphs achieving equality in this bound.

Theorem 5 ([22]). *If G is a connected graph of order n with $\delta(G) \geq 2$, then $\gamma_{R2}(G) \leq \frac{2}{3}n$.*

The graphs achieving this upper bound have been characterized.

Theorem 6 ([22]). *If G is an edge-minimal graph of order n , then $\gamma_{R2}(G) \leq \frac{2}{3}n$.*

The authors provided characterization for edge-minimal graphs G for which $\gamma_{R2}(G) = \frac{2}{3}n$. The authors also established Nordhaus–Gaddum-type results for Roman $\{2\}$ -domination, a type of results that provides extreme values of the sum (or product) of a parameter on a graph G and its complement $\bar{G}$.

Theorem 7 ([22]). *If G is a graph of order $n \geq 3$, then*

$$5 \leq \gamma_{R2}(G) + \gamma_{R2}(\bar{G}) \leq n + 2,$$

and these bounds are tight. Also, if $\gamma_{R2}(G) \leq \gamma_{R2}(\bar{G})$, then $\gamma_{R2}(G) + \gamma_{R2}(\bar{G}) = 5$ if and only if there exists a vertex in G of degree $n - 1$ with a neighbor of degree one in G or with two adjacent neighbors of degree two in G .

The upper bound of the above result can be improved slightly if the graph G has no small components.

Theorem 8 ([22]). *If G is a graph of order $n \geq 16$ and has no component with fewer than three vertices, then*

$$\gamma_{R2}(G) + \gamma_{R2}(\bar{G}) \leq n - 1$$

Let $X \subseteq V(G)$. The set $N(X) := \bigcup_{x \in X} N(x)$ is known as the *open neighborhood* of the set X , while $N_e(X) := N(X) \setminus X$ is the *external neighborhood* of X or the *boundary* of X . Let $\partial(X) := |N_e(X)| - |X|$. The *differential* of a graph G is

$$\partial(G) := \max\{\partial(X) \mid X \subseteq V(G)\}.$$

Hedetniemi presented these ideas in an unpublished manuscript, then Goddard and Henning [23] developed the initial findings on the subject. Several authors, including [24–29], proceeded to develop the subject after that. Since it has been found that studying various forms of domination, for example Roman domination and Roman $\{2\}$ -domination can be achieved through a suitable variation of differentials. The study of differentials in graphs and their variants is currently of significant interest.

For any $x \in X$, the *external private neighborhood of x with respect to X* is defined as

$$P_e(x, X) = \{y \in V(G) \setminus X \mid N(y) \cap X = \{x\}\}.$$

Let $X_w = \{x \in X \mid P_e(x, X) \neq \emptyset\}$. The *strong differential of a set X* is defined to be

$$\partial_s(X) = |N_e(X)| - |X_w|.$$

The *strong differential of G* is

$$\partial_s(G) = \max\{\partial_s(X) \mid X \subseteq V(G)\}.$$

In [30], Martínez and Rodríguez-Velázquez discussed strong differentials in graphs as a new approach to study Roman $\{2\}$ -domination. This approach is helpful in studying the Roman $\{2\}$ -domination number without using functions. The approach derived several new findings on the Roman $\{2\}$ -domination number of a graph, and the most significant one is a Gallai-type theorem, as stated below:

Theorem 9 ([30] (Gallai-type theorem)). *For any graph G of order n ,*

$$\gamma_{R2}(G) + \partial_s(G) = n \tag{1}$$

Using the concept of a strong differential of G , a series of new results is established in [30] which, in combination with Theorem 9, leads to the following conclusion:

- (i) $\gamma_{R2}(G) \geq \gamma(G) + \sigma(G)$;
- (ii) $\gamma_{R2}(G) = \gamma(G)$ if and only if $\gamma_2(G) = \gamma(G)$;
- (iii) if $\delta(G) \geq 2$ then $\gamma_{R2}(G) \leq \frac{1}{2}(n(G) + \gamma(G))$;
- (iv) $\gamma_{R2}(G) \geq \gamma_R(G) - \gamma(G) + 1$;
- (v) $\gamma_{R2}(G) = \frac{2n(G)}{\Delta(G)+2}$ if and only if $\gamma_2(G) = \frac{2n(G)}{\Delta(G)+2}$;
- (vi) if $\delta(G) \geq 2$ then $\gamma_{R2}(G) \leq \alpha(G)$;
- (vii) if G is connected and $\gamma(G) \geq 3$ then $\gamma_{R2}(G) \leq \frac{1}{2}\beta(G)$;
- (viii) if $\delta(G) \geq 1$ then $\gamma_{R2}(G) \geq \gamma_{t2}(G)$;
- (ix) if $\gamma_{t2}(G) = \gamma_2(G)$ then $\gamma_{R2}(G) = \gamma_2(G)$.

Where $\sigma(G)$ is the number of support vertices in G adjacent to two leaves or more, $\gamma_2(G)$ is the 2-domination number (introduced in [31]), $\gamma_{t2}(G)$ is the semitotal domination number (introduced in [32]), $\alpha(G)$ is the independent number, and $\beta(G)$ is the vertex cover number of G .

It is also shown that the problem of finding the strong differential of G is NP-hard.

3.2. Special Classes of Graphs

In this subsection, we survey the results related to special classes of graphs.

Corollary 1 ([1]). Let P_n be a path of order n , and let C_n be a cycle of order n . Then, $\gamma_{R2}(P_n) = \lceil \frac{n+1}{2} \rceil$ and $\gamma_{R2}(C_n) = \lceil \frac{n}{2} \rceil$.

In [33], it was shown if T is a tree of order $n \geq 2$, then $\gamma_{R2}(T) \geq \gamma(T) + 1$. Trees satisfying $\gamma_{R2}(T) = \gamma(T) + 1$, and trees satisfying $\gamma_{R2}(T) = 2\gamma(T)$ are characterized in [7]. The authors in [34] provided a constructive characterization for trees T satisfying $\gamma_{R2}(T) = \gamma_R(T)$. In [22], the authors proved that $\gamma_{R2}(T) \leq \frac{3}{4}n$, where T is a tree of order $n \geq 3$, and they characterized trees achieving equality in this bound.

In [35], the connected graphs G for which $\gamma_{R2}(G) \in \{2, 3, 4, n-2, n-1, n\}$ are determined. Furthermore, the Roman $\{2\}$ -domination number of the joining of two graphs and the corona product of two graphs is determined. The authors also gave a sharp upper bound of $\gamma_{R2}(G)$, where G is the Cartesian product of two graphs, and the exact value of $\gamma_{R2}(G)$ for some grid graphs G .

In [36], Roman $\{2\}$ -domination number for the Cartesian product of the graphs $C_n \square C_m$, where $m = 3, 4$ is given.

Theorem 10 ([36]).

(i) Let n be a positive integer greater than 2. Then,

$$\gamma_{R2}(C_n \square C_3) = \begin{cases} n, & n \equiv 0 \pmod{3}, \\ n+1, & \text{otherwise.} \end{cases}$$

(ii) Let n be a positive integer greater than 3. Then,

$$\gamma_{R2}(C_n \square C_4) = \begin{cases} \lceil \frac{3n}{2} \rceil, & n \equiv 0, 1, 3, 4, 5 \pmod{8}, \\ \lceil \frac{3n}{2} \rceil + 1, & \text{otherwise.} \end{cases}$$

Theorem 11 ([36]). Let n be a positive integer greater than 4. Then, $\gamma_{R2}(C_n \square C_5) \geq 2n$

It was proven in [37] that $\gamma_{R2}(C_n \square C_5) \leq 2n$, where $n \geq 3$. So, from this fact, Theorem 1 and Theorem 11, $\gamma_{R2}(C_n \square C_5) = 2n$ if $n \geq 5$. In [38], the authors studied the Roman $\{2\}$ -domination number of the Cartesian products of cycles $C_n \square C_m$ for $m \geq 6$. For $n \equiv 0 \pmod{3}$, $m \equiv 0 \pmod{3}$, $\gamma_{R2}(C_n \square C_m) = \frac{nm}{3}$. For $m \not\equiv 0 \pmod{3}$, $m \not\equiv 0 \pmod{3}$, $\lceil \frac{nm}{3} \rceil \leq \gamma_{R2}(C_n \square C_m) \leq \lfloor \frac{2nm+n+2m+1}{6} \rfloor$.

In [12], the authors studied Roman $\{2\}$ -domination of the Cartesian product of circles and paths. They determined the exact values of the Roman $\{2\}$ -domination number of $C_n \square P_3$ and $C_3 \square P_m$. Also, they found some bounds on the Roman $\{2\}$ -domination number of $C_n \square P_m$ for $n, m \geq 4$.

Theorem 12 ([12]).

(i) If $G = C_n \square P_3$, then

$$\gamma_{R2}(G) = \begin{cases} \lceil \frac{5}{4}n \rceil, & \text{if } n \equiv 0, 1, 2, 5, 6 \pmod{8}, \\ \lceil \frac{5}{4}n \rceil + 1, & \text{if } n \equiv 3, 4, 7 \pmod{8}. \end{cases}$$

(ii) If $G = C_3 \square P_m$, then

$$\gamma_{R2}(G) = \begin{cases} m + 1, & \text{if } m \leq 3, \\ m + 2, & \text{if } m \geq 4. \end{cases}$$

(iii) If $G = C_n \square P_m$, $n, m \geq 4$, then

$$\gamma_{R2}(G) \geq \frac{mn}{3} + \frac{n}{9}.$$

For the generalized Petersen graph $P(n, 3)$, it was proven in [9] that $\gamma_{R2}(P(n, 3)) = \lceil \frac{4n}{5} \rceil$ if $n \equiv 0, 4 \pmod{5}$ or $n = 8$, and $\gamma_{R2}(P(n, 3)) = \lceil \frac{4n}{5} \rceil + 1$, otherwise. In [39], it was proven that if G is a threshold graph with k components, then $\gamma_{R2}(G) = k + 1$. The authors also proved that if $G = K_{r,s}$ is a complete bipartite graph with $r \leq s$, then $\gamma_{R2}(G) = 2$ if $r = 1$, $\gamma_{R2}(G) = 3$ if $r = 2$, and $\gamma_{R2}(G) = 4$ if $r \geq 3$. In addition, they determined the exact value of $\gamma_{R2}(G)$, where G is a connected chain graph different than $K_{r,s}$.

In [40], Roman $\{2\}$ -domination was used as a parameter to prove a conjecture raised in 2006 [41] for some types of graphs. The authors of [41] were studying the $\{k\}$ -domination number of graphs G , denoted by $\gamma_{\{k\}}(G)$, and asked if there is an integer $k \geq 2$ such that $\gamma_{\{k\}}(H_1 \square H_2) \geq \gamma(H_1)\gamma(H_2)$. In [40], it was proved that if H_1 is any graph, and H_2 is a claw-free graph, then $\gamma_{\{2\}}(H_1 \square H_2) \geq \gamma_{R2}(H_1 \square H_2) \geq \gamma(H_1)\gamma(H_2)$, which indicated that the conjecture is true for every integer $k \geq 2$.

In [42], Roman $\{2\}$ -domination is studied for rooted product graphs. Klavžar and Milutinović [43] introduce the graph $S(K_n, t)$, $t \geq 1$ and later in [44] named as a Sierpiński graph. In [45], the Roman $\{2\}$ -domination number for Sierpiński graphs was determined. A generalized Sierpiński graph $S(G, t)$ of a graph G is introduced in [46]. In [47], Varghese et al. obtained the exact values of the Roman $\{2\}$ -domination number of $S(G, 2)$, for $G \in \{C_n, K_{1,q}, K_{2,q}, B_{m,l}\}$, where $B_{m,n}$ is a bistar graph with $m, l \geq 3$. They also found the exact value of the perfect Roman $\{2\}$ -domination number and the double Roman domination number of $S(G, 2)$ for the same set of G . Given our interests of this section, we noted only the results related to Roman $\{2\}$ -domination number.

Theorem 13 ([47]).

- (i) $\gamma_{R2}(S(C_n, 2)) = n \lceil \frac{n}{2} \rceil$, for $n \geq 4$.
- (ii) $\gamma_{R2}(S(K_{1,q}, 2)) = 2q + 1$, for $q \geq 3$.
- (iii) $\gamma_{R2}(S(K_{2,q}, 2)) = 2(q + 2)$, for $q \geq 3$.
- (iv) $\gamma_{R2}(B_{m,n}, 2) = 4(m + n + 1)$, $m, n \geq 3$.

The effect of the removal of the vertex and edges on the Roman $\{2\}$ -domination number were studied by Nazari–Moghaddam [48]. They also discussed the Roman $\{2\}$ -domination numbers of the Cartesian product of some graphs.

Theorem 14 ([48]). If $m, n \geq 1$, then $\gamma_{R2}(C_{5m} \square C_{5n}) = 10mn$.

Let $D = (V, A)$ be a finite simple digraph with vertex set $V = V(D)$ and arc set $A = A(D)$. The maximum out-degree and maximum in-degree of D are denoted by $\Delta^+(D)$ and $\Delta^-(D)$. A function $f : V(D) \rightarrow \{0, 1, 2\}$ is called a *Roman $\{2\}$ -dominating function* on D if every $v \in V_0^f$ has at least two in-neighbors in V_1^f or at least one in-neighbor in V_2^f . The *Roman $\{2\}$ -domination number* of D (also called the Italian domination number) is the minimum weight of a Roman $\{2\}$ -dominating function on D . L. Volkmann [10] studied the Roman $\{2\}$ -domination number of a digraph D and proved the following results:

Theorem 15 ([10]). *Let D be a digraph of order n . Then,*

- (i) $\gamma_{R2}(D) \geq \left\lceil \frac{2n}{2+\Delta^+(D)} \right\rceil$;
- (ii) $\gamma_{R2}(D) \leq n$, and $\gamma_{R2}(D) = n$ if and only if $\Delta^+(D), \Delta^-(D) \leq 1$.

Theorem 16 ([10]). *If D is directed path or a directed cycle of order n , then $\gamma_{R2}(D) = n$.*

Using Theorem 15 and Theorem 16, K. Kim [13] determined the exact values of $\gamma_{R2}(C_m \square C_n)$ and $\gamma_{R2}(C_m \otimes C_n)$, where $C_m \otimes C_n$ denotes the strong product of the directed cycles C_m and C_n .

Theorem 17 ([13]).

- (i) *If $m = 2r$ and $n = 2s$ for some positive integer r and s , then $\gamma_{R2}(C_m \square C_n) = mn/2$.*
- (ii) *For an odd integer $n \geq 3$, $\gamma_{R2}(C_2 \square C_n) = n + 1$.*
- (iii) *For an odd integer $n \geq 3$, $\gamma_{R2}(C_3 \square C_n) = 2n$.*
- (iv) *For positive integer $m, n \geq 2$, $\gamma_{R2}(C_3 \otimes C_n) = \lceil \frac{mn}{2} \rceil$.*

The following conjecture is also given in [13].

Conjecture 1. *For an odd integer n , $\gamma_{R2}(C_4 \square C_n) = 2n + 2$.*

The investigation of the Roman $\{2\}$ -domination numbers of the Cartesian product of directed cycles was completed in [14].

3.3. Algorithmic Complexity

This subsection is devoted for algorithmic complexity of Roman $\{2\}$ -domination problems.

It was shown in [17] that finding the 2-rainbow domination number for trees can be achieved in linear time; therefore, the Roman $\{2\}$ -domination number for trees can also be found in linear time. In [1], it was shown that the Roman $\{2\}$ -domination problem is NP-complete for bipartite graphs. In [49], the authors proved that the Roman $\{2\}$ -domination problem is NP-complete for split graphs. The authors also provided a linear time algorithm computing the Roman $\{2\}$ -domination number for block graphs.

In [50], it was shown that the Roman $\{2\}$ -domination problem is NP-complete when restricted to the planar graph. The authors also gave a linear algorithm to determine $\gamma_{R2}(G)$ of unicyclic graphs G . In [39], it was proven that the Roman $\{2\}$ -domination problem is NP-complete for bisplit graphs, for comb bipartite graphs, and for star convex bipartite graphs.

Fernández and Leoni [51] studied NP-complete instances of the Roman $\{2\}$ -domination problem on several graphs, namely, chordal graphs, bipartite planar graphs, chordal bipartite graphs, bipartite with maximum degree 3 graphs, and many more. They also established an algorithm that is independent of k -rainbow domination and computes the Roman $\{2\}$ -domination number on caterpillars, which is a subclass of trees.

4. Roman $\{2\}$ -Domination Variants

This subsection is devoted to Roman $\{2\}$ -domination variants. Below are 15 variants seen in the literature with their main results and open problems.

4.1. Perfect Roman $\{2\}$ -Domination

Let $G = (V, E)$ be a graph. A function $f : V \rightarrow \{0, 1, 2\}$ is called a perfect Roman $\{2\}$ -dominating function on G (PR2DF) if for every $v \in V_0^f$, $\sum_{u \in N(v)} f(u) = 2$. The perfect Roman $\{2\}$ -domination number of G , denoted by $\gamma_{R2}^p(G)$, is the minimum weight of a PR2DF on G . This variant of Roman $\{2\}$ -domination was introduced in [52], and it was called perfect Italian domination. The authors in [52] were interested in finding the best possible constant K_C such that $\gamma_{R2}^p(G) \leq K_C n$, where G is in the class $\mathcal{C}$ of graphs. The authors provided K_C for the class of connected bipartite graphs and for the class of trees.

Lemma 1 ([52]). *If $\mathcal{C}$ is the class of connected bipartite graphs, then $K_C = 1$.*

The following was the main result in [52]:

Theorem 18 ([52]). *If $\mathcal{C}$ is the class of trees with at least three vertices, then $K_C = \frac{4}{5}$.*

For every positive integer $n \equiv 0 \pmod{5}$, the authors constructed a tree T of order n achieving this bound, i.e., $\gamma_{R2}^p(T) = \frac{4n}{5}$. The authors raised the following open problem:

Problem 4. *Find K_C for other classes of graphs, for example, the class of graphs with large girth, the class of regular graphs, or the class of planar graphs.*

In [53], it was shown that $K_C = 1$ for the class of planar graphs, $K_C = \frac{2}{3}$ for the class of cubic graphs, and $K_C = 1$ for the class of split graphs. Recently, a new upper bound for the perfect Roman $\{2\}$ -domination number of trees was given [54]; this new upper bound improved the bound in Theorem 18, especially for trees with a high number of leaves compared to the number of support vertices.

Theorem 19 ([54]). *Let T be a tree of order at least $n \geq 3$ with l leaves and s support vertices. Then,*

$$\gamma_{R2}^p(T) \leq \frac{4n - l + 2s - 1}{5}.$$

In [45], the perfect Roman $\{2\}$ -domination number for Sierpiński graphs was obtained, and in [47], the perfect Roman $\{2\}$ -domination number was determined for the generalized Sierpiński graphs $S(G, 2)$ for particular types of graphs G . In [55], it was proven that if G is a cograph, then $\gamma_{R2}^p(G) \in \{1, 2, 3, 4, |V(G)|\}$. In [56], an upper bound for the perfect Roman $\{2\}$ -domination number of the Cartesian product of any two graphs was provided, and the exact perfect Roman $\{2\}$ -domination number of the Cartesian product of some graphs was obtained. The authors also studied the relation between the perfect Roman $\{2\}$ -domination number and perfect domination, as well as the relation between the perfect Roman $\{2\}$ -domination number and the Roman domination number. In [57], it was shown that the decision problem associated with the perfect Roman $\{2\}$ -domination is an NP-complete for chordal graphs, and the authors presented a linear time algorithm to find $\gamma_{R2}^p(G)$ for block graphs and series-parallel graphs. The concept of perfect Roman $\{2\}$ -domination was generalized in [58] to perfect Roman $\{k\}$ -domination for any $k \in \mathbb{Z}^+$.

4.2. Total Roman $\{2\}$ -Domination

Let $G = (V, E)$ be a graph. The function $f : V \rightarrow \{0, 1, 2\}$ is a total Roman $\{2\}$ -dominating function on G if

- (i) every vertex $v \in V_0^f$ satisfies that $\sum_{u \in N(v)} f(u) \geq 2$, and
- (ii) every vertex $x \in V_1^f \cup V_2^f$ is adjacent to at least one vertex $y \in V_1^f \cup V_2^f$.

The total Roman $\{2\}$ -dominating number, denoted by γ_{tR2} , is the minimum weight among all total Roman $\{2\}$ -dominating functions in G . García et al. [59] introduced the

concept of total Roman $\{2\}$ -domination and investigated its combinatorial and computational properties. To have a complete picture of the total Roman $\{2\}$ -domination, we present below a few results from [59].

Proposition 2 ([59]). *For any graph G without isolated vertices, the following inequalities hold:*

- (i) $\gamma_t(G) \leq \gamma_{tR2}(G) \leq \gamma_{tR}(G) \leq 2\gamma_t(G)$.
- (ii) $\gamma_{R2}(G) \leq \gamma_{tR2}(G) \leq \gamma_{\times 2}(G)$.

The numbers $\gamma_t(G)$, $\gamma_{tR}(G)$ and $\gamma_{\times 2}(G)$ are the total domination number (introduced in [60]), total Roman domination number (introduced in [61]), and double domination number (introduced in [62]), respectively. The inequality $\gamma_{tR}(G) \leq 2\gamma_t(G)$ in part (i) is obtained in [63].

Remark 1 ([59]). *For any graph G , the following statements are equivalent:*

- (i) $\gamma_{tR2}(G) = \gamma_t(G)$.
- (ii) $\gamma_{\times 2}(G) = \gamma_t(G)$.

In continuation, the following results show a simple relation between the total Roman $\{2\}$ -domination with the domination number and the total domination number:

Theorem 20 ([59]). *For any graph G without isolated vertices,*

$$\gamma_{tR2}(G) \leq \gamma_t(G) + \gamma(G).$$

Since for any graph G , $\gamma(G) \leq \gamma_t(G)$, Theorem 20 improves the last inequality of Proposition 2. Due to the well-known inequality $\gamma(G) \leq \gamma_t(G)$, Theorem 20 implies $\gamma_{tR2}(G) \leq 3\gamma(G)$. The upper bound of Theorem 20 is sharp.

The following results gives us an equivalent condition for the graphs G which satisfy the equality $\gamma_{tR2}(G) = 2\gamma_t(G)$:

Theorem 21 ([59]). *For a graph G , $\gamma_{tR2}(G) = 2\gamma_t(G)$ if and only if $\gamma_t(G) = \gamma(G)$.*

The next result demonstrates a relation of total Roman $\{2\}$ -domination with Roman $\{2\}$ -domination and total domination.

Theorem 22 ([59]).

- (i) *For any graph G without isolated vertices, $\gamma_{tR2}(G) \leq \gamma_{R2}(G) + \gamma(G)$. The bound is sharp and holds for the star graph $K_{1,n-1}$ when $n \geq 3$.*
- (ii) *For any graph G of order n and $\delta(G) \geq 2$,*

$$\gamma_{tR2}(G) \leq \left\lfloor \frac{\gamma_t(G) + n}{2} \right\rfloor.$$

Tightness of bound can be validated through the Cartesian product graph $P_2 \square P_3$.

The following results for the path, cycle, and Hamiltonian graph can be summarized from [59]:

- (i) For $n \geq 2$,

$$\gamma_{tR2}(P_n) = \begin{cases} 2\lceil \frac{n}{3} \rceil + 1, & \text{if } n \equiv 0(\text{mod}3), \\ 2\lceil \frac{n}{3} \rceil, & \text{Otherwise.} \end{cases}$$

- (ii) For $n \geq 2$, $\gamma_{tR2}(C_n) = 2\lceil \frac{2n}{3} \rceil$.
- (iii) If the graph G is a Hamiltonian graph of order n , then $\gamma_{tR2}(G) \leq 2\lceil \frac{n}{3} \rceil$.
- (iv) If the graph G of order n has a Hamiltonian path, then $\gamma_{tR2}(G) \leq 2\lceil \frac{n}{3} \rceil + 1$.

A detailed investigation is conducted to characterize trees T satisfying $\gamma_{tR2}(T) = \gamma_{tR}(T)$ [59].

The work [64–67] provided additional development on the total Roman $\{2\}$ -domination and the total Roman $\{2\}$ -dominating function. The article [64] provided a number of bounds on the total Roman $\{2\}$ -domination number, one of those bounds is that $\gamma_{tR2}(G) \leq 3\gamma(G)$ for every nontrivial connected graph G , and they demonstrated the NP-completeness of the decision problem related to $\gamma_{tR2}(G)$ for both bipartite and chordal graphs. They also proved that for bounded clique-width graphs (which include trees), this parameter can be computed in linear time. The following two open problems are stated in [64]:

Problem 5. For a given graph G , is the problem of deciding whether $\gamma_{tR2}(G) = 3\gamma(G)$ NP-hard?

Problem 6. Characterize all graphs G such that $\gamma_{tR2}(G) = 3\gamma(G)$.

A necessary condition for nontrivial connected graphs G with $\gamma_{tR2}(G) = 3\gamma(G)$ is presented in [65], which also characterizes graphs that are diameter-2 or trees. This provides a partial solution to Problem 6. Necessary and sufficient conditions for nontrivial connected graphs for which $\gamma_{tR2}(G) = \gamma_t(G)$ are also established in [65] and characterize those graphs that are $\{C_3, C_6\}$ -free or block.

After surveying work related to Roman $\{2\}$ -domination and its variants, we have not seen a variant that combines both perfect and total Roman $\{2\}$ -domination. For work combining perfect and total Roman domination, we refer the reader to [68].

4.3. Independent Roman $\{2\}$ -Domination

A Roman $\{2\}$ -domination function is independent if the set of vertices having positive function values is an independent set. The minimum weight of an independent Roman $\{2\}$ -dominating function on G is called the independent Roman $\{2\}$ -domination number and it is denoted by $i_{R2}(G)$. The independent Roman $\{2\}$ -domination was studied in [69]. The authors showed that the decision problem associated with independent Roman $\{2\}$ -domination is NP-complete even when restricted to bipartite graphs. Also, they showed that for every graph G of order n , $0 \leq i_{r2}(G) - i_{R2}(G) \leq \frac{n}{5}$ and $0 \leq i_R(G) - i_{R2}(G) \leq \frac{n}{4}$, and those upper bounds are sharp, where $i_{r2}(G)$ and $i_R(G)$ are the independent 2-rainbow domination and independent Roman domination numbers, respectively. Also, they showed that $i_{R2}(T) = i_{r2}(T)$, where T is any tree. The authors proposed the following problems:

Problem 7. Can you design a linear algorithm for computing the value of $i_{R2}(T)$ for any tree T ?

Problem 8. Characterize the graphs (or at least the trees) G for which $\gamma_{R2}(G) = i_{R2}(G)$.

Problem 9. Characterize the graphs G for which $i_{R2}(G) = i_R(G)$.

Problem 10. Characterize the graphs G for which $i_{R2}(G) = i_{r2}(G)$.

Two of the above open questions were answered by [70]. The first problem was answered independently by [71–73].

In [74], the authors found that the independent Roman $\{2\}$ -domination problem is linear-time solvable for bounded tree-width graphs, chain graphs, and threshold graphs, a subclass of split graphs. Also, they found that the independent Roman $\{2\}$ -domination problem is APX-hard for graphs with bounded degree 4 and is NP complete for star convex bipartite graphs, chordal graphs, dually chordal graphs, and tree convex bipartite graphs.

In [72], the authors proved that the independent Roman $\{2\}$ -domination problem is NP-complete when restricted to planar graphs.

In [73], they authors proved that $i_{R2}(T) \geq \gamma(T) + 1$ for any nontrivial tree T and characterized all trees with $i_{R2}(T) = \gamma(T) + 1$.

In [75], the authors initiated the study of independent Roman $\{2\}$ -bondage. The independent Roman $\{2\}$ -bondage number $b_{iR2}(G)$ of a graph G with at least one component of order at least three is the least number of edges whose removal increases the independent Roman $\{2\}$ -domination number. They found that the independent Roman $\{2\}$ -bondage problem is NP-hard for arbitrary graphs. Bounds on $b_{iR2}(G)$ are given for some graphs; in particular, they found that $b_{iR2}(T) \leq 2$, where T is a tree of order at least three.

4.4. Roman $\{2\}$ -Reinforcement and Roman $\{2\}$ -Bondage

The Roman $\{2\}$ -reinforcement number of a graph G , denoted by $r_{R2}(G)$, is the minimum number of edges need to be added to the graph to decrease the Roman $\{2\}$ -domination number. This notion was initiated in [76], and the authors called it Italian reinforcement domination. Note that if $\gamma_{R2}(G) \in \{1, 2\}$, then no set of edges can be added to decrease the Roman $\{2\}$ -domination number [76]. Thus, when discussing the Roman $\{2\}$ -reinforcement number, we assume that $\gamma_{R2}(G)$ is at least 3. The authors provided a characterization for graphs G for which $r_{R2}(G) = 1$, and they also obtained some sharp upper bounds of $r_{R2}(G)$ for graphs G with $\gamma_{R2}(G) \geq 3$.

The Roman $\{2\}$ -reinforcement number is determined for well-known classes of graphs, namely, paths, cycles, complete multipartite graphs, and the Cartesian product graph $P_2 \square P_n$ [76].

We propose the following problem:

Problem 11. Study the complexity of determining $r_{R2}(G)$.

The notion of Roman $\{2\}$ -reinforcement number was extended to digraphs [77]. The authors presented some bounds for the Roman $\{2\}$ -reinforcement number of digraphs, and determined the Roman $\{2\}$ -reinforcement number for some classes of digraphs. In [78], sharp upper bounds for the Roman $\{2\}$ -reinforcement number of digraphs were given, and the exact value of the Roman $\{2\}$ -reinforcement number for the Cartesian product of some directed cycles and paths were determined.

The Roman $\{2\}$ -bondage number of G , denoted by $b_{R2}(G)$, is the minimum number of edges that needs to be removed from the graph to increase the Roman $\{2\}$ -domination number. This notion was initiated in [79]. The authors proved that the Roman $\{2\}$ -bondage problem is NP-hard; they also presented an upper bound for the Roman $\{2\}$ -bondage number. The authors proved that the Roman $\{2\}$ -bondage number for planar graphs is at most 15, and they concluded the paper by determining the exact value of Roman $\{2\}$ -bondage number for some classes of graphs.

The notion of Roman $\{2\}$ -bondage number was extended to digraphs [77]. The authors presented some bounds for the Roman $\{2\}$ -bondage number of digraphs, and determined the Roman $\{2\}$ -bondage number for some classes of digraphs.

4.5. Total Roman $\{2\}$ -Reinforcement Number

The total Roman $\{2\}$ -reinforcement number of a graph G , denoted by $r_{tR2}(G)$, is the minimum number of edges that needs to be added to the graph to decrease the total Roman $\{2\}$ -domination number. The total Roman $\{2\}$ -reinforcement number was introduced in [80]. Some properties of $r_{tR2}(G)$, sharp upper bounds, and the exact value of $r_{tR2}(G)$ for paths and complete multipartite graphs were given in [80].

We remark that if $\gamma_{tR2}(G) = 2$, then adding edges does not decrease the total Roman $\{2\}$ -domination number; therefore, when discussing the total Roman $\{2\}$ -reinforcement number, it was always assumed that $\gamma_{tR2}(G) \geq 3$. The authors of [80] showed that if $\gamma_{tR2}(G) \geq 3$, then $r_{tR2}(G) \leq \Delta(G) + 1$, and if $\gamma_{tR2}(G) \geq 4$, then $r_{tR2}(G) \leq n - \Delta(G) - 1$, which implies that $r_{tR2}(G) \leq \lceil \frac{n}{2} \rceil$ for every graph G of order n with $\gamma_{tR2}(G) \geq 4$. The authors also characterized graphs G for which $r_{tR2}(G) = 1$. For paths P_n of order at least 3, $r_{tR2}(P_n) = 1$ if $n \equiv 0, 1 \pmod{3}$, and $r_{tR2}(P_n) = 2$, otherwise. We finish this subsection with the following upper bound:

Theorem 23 ([80]). Let G be a graph of order n with $\gamma_{tR2}(G) \geq 4$. Then,

$$r_{tR2}(G) \leq n - \Delta(G) - \left\lfloor \frac{\gamma_{tR2}(G)}{2} \right\rfloor + 1.$$

4.6. Restrained Roman $\{2\}$ -Domination

A restrained Roman $\{2\}$ -dominating function of a graph G is a Roman $\{2\}$ -dominating function with the following extra condition is met: the subgraph induced by the set $\{v \in V(G) | f(v) = 0\}$ contains no isolated vertex. The restrained Roman $\{2\}$ -domination number of G , $\gamma_{rR2}(G)$, is the minimum weight of a restrained Roman $\{2\}$ -dominating function on G . The restrained Roman $\{2\}$ -domination (also called restrained Italian domination) was introduced in [81]; the authors proved that the restrained Roman $\{2\}$ -domination problem is NP-hard for bipartite graphs, chordal graphs, and planar graphs G with $\Delta(G) \leq 5$, and they determined a sharp upper bound of $\gamma_{rR2}(T)$ for tree graphs T and characterized trees achieving this upper bound. The authors proposed some open problems:

Problem 12. Investigate the complexity of the restrained Roman $\{2\}$ -domination problem for other classes of graphs, for example, trees.

Problem 13. It was shown that $\gamma_{re}(G) \leq \gamma_{rR2}(G) \leq 2\gamma_{re}(G)$ for any graph G , where $\gamma_{re}(G)$ is the restrained domination number of G . Characterize all graphs G such that $\gamma_{re}(G) = \gamma_{rR2}(G)$. Characterize all graphs G such that $\gamma_{rR2}(G) = 2\gamma_{re}(G)$.

Problem 14. Determine sharp bounds of $\gamma_{rR2}(G)$ for other classes of graphs.

We remark that in [81], the restrained domination number of G was denoted by $\gamma_r(G)$, but we created $\gamma_{re}(G)$ as a notation of the restrained domination number since $\gamma_r(G)$ is used in this paper as the notation of weak Roman domination number.

In [82], the author characterized all trees T such that $\gamma_{re}(T) = \gamma_{rR2}(T)$, and characterized all trees T such that $\gamma_{rR2}(T) = 2\gamma_{re}(T)$. In [83], some Nordhaus–Gaddum-type results, as well as sharp bounds, of restrained Roman $\{2\}$ -domination number were presented.

In [84], the total restrained Roman $\{2\}$ -domination number was defined; and restrained and total restrained Roman $\{2\}$ -domination numbers for some classes of graphs were provided.

In [85,86], the authors defined the restrained Italian reinforcement number and restrained Italian bondage number; they proved that the decision problems associated with these variants are NP-hard, and some properties and sharp bounds of these variants were given.

4.7. Secure Roman $\{2\}$ -Domination

The notion of secure Roman $\{2\}$ -domination was introduced and studied in [87] as a special case of secure w -domination, which was introduced in [88]. Secure Roman $\{2\}$ -domination was called secure Italian domination in [88]. Let f be a Roman $\{2\}$ -dominating function and (u, v) be a pair of adjacent vertices in $V(G)$ such that $f(u) = 0$ and $f(v) \geq 1$. The function $f_{v \rightarrow u}$ of $V(G)$ is defined by setting $f_{v \rightarrow u}(u) = 1$, $f_{v \rightarrow u}(v) = f(v) - 1$ and $f_{v \rightarrow u}(a) = f(a)$, otherwise. The secure Roman $\{2\}$ -dominating function is a Roman $\{2\}$ -dominating function f satisfying that every vertex $u \in V(G)$ with $f(u) = 0$ has a neighbor v with $f(v) \geq 1$ such that the function $f_{v \rightarrow u}$ is a Roman $\{2\}$ -dominating function. The secure Roman $\{2\}$ -domination number, denoted by $\gamma_{R2}^s(G)$, is the minimum weight of a secure Roman $\{2\}$ -dominating function on G .

In [87], the authors proved that the secure Roman $\{2\}$ -domination problem is NP-hard, they obtained general bounds on secure Roman $\{2\}$ -domination number, and they studied secure Roman $\{2\}$ -domination number for particular classes of graphs.

As mentioned earlier, secure Roman $\{2\}$ -domination is a special case of secure w -domination. Let $w = (w_0, w_1, \dots, w_t)$ be a vector of non-negative integers w_i with

$w_0 > 0$. The function $f : V(G) \rightarrow \{0, 1, \dots, t\}$ is called a w -dominating function if for every $v \in V(G)$ with $f(v) = i$, the sum $\sum_{u \in N(v)} f(u) \geq w_i$. Let f be a w -dominating function and (u, v) be a pair of adjacent vertices in $V(G)$ such that $f(u) = 0$ and $f(v) \geq 1$. The function $f_{v \rightarrow u}$ on $V(G)$ is defined by setting $f_{v \rightarrow u}(u) = 1$, $f_{v \rightarrow u}(v) = f(v) - 1$ and $f_{v \rightarrow u}(a) = f(a)$, otherwise. The secure w -dominating function is a w -dominating function f satisfying that every vertex $u \in V(G)$ with $f(u) = 0$ has a neighbor v with $f(v) \geq 1$ such that the function $f_{v \rightarrow u}$ is a w -dominating function. The secure w -domination number is the minimum weight of a secure w -dominating function on G . If $w = (2, 0, 0)$, then the secure w -domination number is precisely the secure Roman $\{2\}$ -domination number. If $w = (1, 0)$, then the secure w -domination number is the secure domination number, which was introduced in [89] and investigated further in [90–93].

4.8. Quasi-Total Roman $\{2\}$ -Domination

A Roman $\{2\}$ -dominating function f on a graph G is called a quasi-total Roman $\{2\}$ -dominating function if every $v \in V_2^f$ has a neighbor in $V_1^f \cup V_2^f$. The quasi-total Roman $\{2\}$ -domination number, denoted by $\gamma_{R2^*}(G)$, is the minimum weight of a quasi-total Roman $\{2\}$ -dominating function on G [94]. Quasi-total Roman $\{2\}$ -domination was called quasi-total Italian domination in [94].

The quasi-total strong differential of G is defined in [94] as follows:

$$\partial_{s^*}(G) = \max\{\partial_s(X) \mid X \subseteq V(G) \text{ and } X_w \subseteq N(X)\}.$$

The quasi-total strong differential is shown to be closely correlated with a number of graph parameters, such as the quasi-total Roman $\{2\}$ -domination number, the semitotal domination number, the vertex cover number, the domination number, the total domination number, and the 2-domination number [94].

For completeness, we summarized prime findings of [94] in the theorem below.

Theorem 24 ([94]). Let G be a graph of order n . Then, the following statements hold:

- (i) $\partial_s(G) - \gamma(G) \leq \partial_{s^*}(G) \leq \partial_s(G)$.
- (ii) $n - \min\{3\gamma(G), \gamma_2(G)\} \leq \partial_{s^*}(G)$.
- (iii) If $\Delta(G) \leq 3$ or G is a claw-free graph, then $\partial_{s^*}(G) = n - \gamma_2(G)$.
- (iv) For any integer $n \geq 3$, $\partial_{s^*}(P_n) = \lfloor \frac{n-1}{2} \rfloor$ and $\partial_{s^*}(C_n) = \lfloor \frac{n}{2} \rfloor$.
- (v) If $\delta(G) \geq 3$, then $\partial_{s^*}(G) = \frac{n}{2}$.
- (vi) If $\delta(G) = 2$, then $\partial_{s^*}(G) = \frac{n}{3}$.
- (vii) If G is a graph with no isolated vertex, then
 - (a) $\partial_{s^*}(G) \leq n - \gamma_{t2}(G)$,
 - (b) $\partial_{s^*}(G) = n - \gamma_{t2}(G)$ if and only if $\gamma_{t2}(G) = \gamma_2(G)$, and
 - (c) $\partial_{s^*}(G) \geq n - \gamma_t(G) - \gamma(G)$.
- (viii) For any graph G with every component of order at least three, then
 - (a) $\partial_{s^*}(G) \geq \alpha(G) - |\mathcal{S}(G)| - |\mathcal{S}_s(G)|$,
 - (b) $\partial_{s^*}(G) \geq n - \beta(G) - |\mathcal{S}_s(G)|$, and
 - (c) $\partial_{s^*}(G) \geq \lceil \frac{1}{2}(n - \gamma(G) + |\mathcal{L}(G)| - 2|\mathcal{S}(G)| - 2|\mathcal{S}_s(G)|) \rceil$.
- (ix) Gallai-type theorem: For any graph G , $\gamma_{R2^*}(G) + \partial_{s^*}(G) = n$.

Where $\mathcal{L}(G)$, $\mathcal{S}(G)$, and $\mathcal{S}_s(G)$ denote the number of leaves, support, and strong support vertices of G , respectively.

In the end, article [94] concluded that the problem of determining the quasi-total strong differential of a graph is NP-hard. The following open problems are also highlighted in the article:

Problem 15. Find Nordhaus–Gaddum-type relations for $\partial_{s^*}(G)$.

Problem 16. Characterize all graphs such that $\partial_{s^*}(G) = \alpha(G)$ and $\partial_{s^*}(G) = \gamma(G)$.

Problem 17. Develop polynomial-time algorithms for basic graph families or develop heuristics to maximize the accuracy of parameter estimation for any graph, for instance, determining whether $\partial_{s^*}(G)$ is an NP-hard optimization problem.

Problem 18. Examine the quasi-total strong differential of product graphs and attempt to formulate this invariant in terms of various graph parameters.

4.9. Signed Roman $\{2\}$ -Domination

A signed Roman $\{2\}$ -dominating function (SR2DF) on a graph G is a function $f : V \rightarrow \{-1, 1, 2\}$ with the property that for every vertex $v \in V$, $f(N[v]) \geq 1$. The signed Roman $\{2\}$ -domination number of a graph G , denoted by $\gamma_{sR2}(G)$ is defined as

$$\gamma_{sR2}(G) = \min\{w(f) \mid f \text{ is an SR2DF on } G\}.$$

The concept of signed Roman $\{2\}$ -domination was first introduced in [95], and the authors called it signed Italian domination. In addition to calculating the signed Roman $\{2\}$ -domination number of different classes of graphs, the authors also found multiple lower bounds on the signed Roman $\{2\}$ -domination number of a graph. For a graph G without an isolated vertex, it is shown that $\gamma_{sR2}(G) \geq (3n - 4m)/2$, where m is the number of edges in G . All graphs reaching equality in this bound are characterized. They also determined that the sharp bound $\gamma_{sR2}(G) \geq 3\sqrt{n}/2 - n$ exists for a graph G of order $n \geq 2$.

The concept of signed double Roman domination was first introduced and studied in [96], and later, Almulhim [97] relaxed the conditions of signed double Roman domination and introduced signed double Italian domination (SDID). A function $f : V \rightarrow \{-1, 1, 2, 3\}$ on a graph $G = (V, E)$ is called a signed double Italian dominating function (SDIDF) on G if (i) $\sum_{u \in N[v]} f(u) \geq 1$ for every $v \in V$; (ii) for every vertex v with $f(v) = -1$, there exists $A \subseteq N(v)$ such that $\sum_{u \in A} f(u) \geq 3$; and (iii) for every vertex v with $f(v) = 1$, there exists $A \subseteq N(v)$ such that $\sum_{u \in A} f(u) \geq 2$. The minimum weight of an SDIDF on G is known as the signed double Italian domination number of G , and it is denoted by $\gamma_{sdI}(G)$. Tight lower and upper bounds of the $\gamma_{sdI}(G)$ of trees were provided, and the trees witnessing those bounds were characterized, and for some well-known graphs, the precise values of $\gamma_{sdI}(G)$ were computed in [97]. The following two open problems were highlighted in [97]:

Problem 19. Characterize connected graphs G with $|G| \geq 3$, satisfying $\gamma_{sdI}(G) = |G|$.

Problem 20. Give characterizations for graphs G satisfying $\gamma_{sdI}(G) = \gamma_{sdR}(G)$. Here, $\gamma_{sdR}(G)$ is signed double Roman domination number.

The signed total Italian domination number of digraphs $\gamma_{sTI}(D)$ was first studied by Volkmann [98]. The article computes the signed total Italian domination number of a few digraph classes and provides several bounds on $\gamma_{sTI}(D)$. The work of [98] is further generalized by the same author to the signed total Italian k -domination of graphs $\gamma_{sTI}^k(G)$.

4.10. Starred Roman $\{2\}$ -Domination

The concept of starred Roman $\{2\}$ -domination was introduced in [99], and the author called it starred Italian domination. A Roman $\{2\}$ -dominating function f for which V_0^f is not a dominating set of the graph G is known as a starred Roman $\{2\}$ -dominating function. The minimum weight among all starred Roman $\{2\}$ -dominating functions on G is called the starred Roman $\{2\}$ -domination number of G , and it is denoted by $\gamma_{R2}^*(G)$. The author in [99] gave some relationships between starred Roman $\{2\}$ -domination and other types of domination, such as Roman $\{2\}$ -domination, domination, and 2-domination; they also established some tight bounds for starred Roman $\{2\}$ -domination number as well as the

exact value of the starred Roman $\{2\}$ -domination number for some graphs. It was also demonstrated that determining a graph's starred Roman $\{2\}$ -domination number is an NP-hard problem.

To the best of our knowledge, there may not be any other studies on this notion in the literature outside of [99].

The following open problems are listed in [99]:

Problem 21. Can we characterize the families of trees T for which $\gamma_{R2}^*(T) = \gamma_{R2}(T)$ or $\gamma_{R2}^*(T) = \gamma_{R2}(T) + 1$?

Problem 22. Characterize all graphs for which $\gamma_{R2}^*(G) = \gamma(G) + 1$.

Problem 23. Is there a polynomial-time algorithm for finding $\gamma_{R2}^*(G)$ for some specific families of graphs?

4.11. The Roman $\{2\}$ -Domestic Number

A set $\{f_1, f_2, \dots, f_d\}$ of distinct Roman $\{2\}$ -dominating functions on G with the property that $\sum_{i=1}^d f_i(v) \leq 2$ for each $v \in V(G)$ is called a Roman $\{2\}$ -dominating family of functions on G . The Roman $\{2\}$ -domatic number of a graph G is the maximum number of functions in a Roman $\{2\}$ -dominating family and is denoted by $d_{R2}(G)$. The study of Roman $\{2\}$ -domatic number was initiated by Volkmann [100], and they determined the value of $d_{R2}(G)$ for some classes of graph, such as cycles, paths, and complete multipartite graphs, and presented different sharp bounds on $d_{R2}(G)$. We list some findings of [100] in the following theorem.

Theorem 25 ([100]). If G is a graph of order n , then the following statements hold:

- (i) $\gamma_{R2}(G) \cdot d_{R2}(G) \leq 2n$.
- (ii) $d_{R2}(G) = n$ if and only if G is isomorphic to the complete graph K_n .
- (iii) If $n \geq 2$, then $\gamma_{R2}(G) + d_{R2}(G) \leq n + 2$, and the equality holds if and only if $\Delta(G) = 1$ or G is the complete graph.

The article also discussed Nordhaus–Gaddum-type results for the Roman $\{2\}$ -domatic number. For $d_{R2}(G)$, the result can be stated as follows:

Theorem 26 ([100]). Let G be a graph of order n . Then,

$$d_{R2}(G) + d_{R2}(\overline{G}) \leq n + 2,$$

with exception of the cases that G is 4-regular of order 9, 7-regular of order 18, or 16-regular of order 45.

To prove Theorem 26, the author in the article also established the following result for regular graphs:

Theorem 27 ([100]). Let G be a δ -regular graph of order n with $\delta \geq 1$, $n = p(\delta + 2) + r$ with integers $p \geq 1$ and $0 \leq r \leq \delta + 1$. If $1 \leq r \leq (\delta + 2)/2$ or $(\delta + 2)/2 < r \leq \delta + 1$, then $d_{R2}(G) \leq \delta + 1$.

Similar kinds of results were obtained by the same author in [101] for digraphs. Three similar type conjectures were proposed in both papers ([100,101]), and it is anticipated that the exception cases in Theorem 26 can be eliminated, and the result is valid for any graph of order n . Jeremy [102] pointed out that the Theorem 26 exception cases, however, are prompted by the limitations of the proof technique and disprove the cases with multiple results.

First, consider the following conjecture from [100]:

Conjecture 2 ([100]). Let G be a δ -regular graph of order n . then, $d_{R2}(G) \leq \delta + 1$.

Jeremy [102] (Theorem 6) counters that there is an infinite family of δ -regular graphs such that $d_{R2}(G) = \delta + 2$. The second conjecture in [100] eliminates the exceptions in Theorem 26 and is stated as follows:

Conjecture 3 ([100]). If G is a regular graph of order n , then $d_{R2}(G) + d_{R2}(\overline{G}) \leq n + 1$.

The cases of a 4-regular of order 9 and a 7-regular of order 18 were considered independently in [102] (Theorem 2) and proved the following.

- (i) There is one 4-regular graph of order $n = 9$ for which $d_{R2}(G) + d_{R2}(\overline{G}) = n + 3$.
- (ii) There is no 7-regular graph of order $n = 18$ for which $d_{R2}(G) + d_{R2}(\overline{G}) = n + 3$.

In the process of disproving Conjectures 2 and 3 through counter examples, there are several interesting interpretations in [102], and they concluded with the following open problem:

Problem 24. Is there a 28-regular graph G on 45 vertices such that $d_{R2}(G) + d_{R2}(\overline{G}) = n + 3$?

In similar line of study, the restrained Roman domatic number ($d_{rR}(G)$) and restrained Roman $\{2\}$ -domatic number ($d_{rR2}(G)$) of a graph G were studied in [103]. The article presented several sharp bounds of $d_{rR}(G)$ and $d_{rR2}(G)$ and determined these parameters for some well-known graphs.

4.12. The Covering Roman $\{2\}$ -Domination

A function f is called a covering Roman $\{2\}$ -dominating function whenever f is a Roman $\{2\}$ -dominating function, and the set $\{v \in V(G) | f(v) \neq 0\}$ is a vertex cover set. The covering Roman $\{2\}$ -domination number, denoted by $\gamma_{CR2}(G)$, is the minimum weight of a covering Roman $\{2\}$ -dominating function on G . This notion was introduced in [104], and the authors called it covering Italian domination. The authors showed that covering $\{2\}$ -Roman domination is NP-hard even when restricted to some well-known graphs. Also, they found some bounds of $\gamma_{CR2}(G)$ for some well-known graphs. In particular, for any graph G , they found that $\beta(G) \leq \gamma_{CR2}(G) \leq 2\beta(G)$ where $\beta(G)$ is the vertex cover number, that is, the minimum cardinality among all vertex cover sets of G . All graphs for which $\gamma_{CR2}(G) = 2\beta(G)$ were characterized.

4.13. The Global Roman $\{2\}$ -Domination

A Roman $\{2\}$ -dominating function f on a graph G is called a global Roman $\{2\}$ -dominating function on G if f is also a Roman $\{2\}$ -dominating function f on $\overline{G}$. The global Roman $\{2\}$ -domination number of G , denoted by $\gamma_{gR2}(G)$, is the minimum weight of a global Roman $\{2\}$ -dominating function f on G . This notion was introduced and investigated in [105], and the authors called it global Italian domination. The authors proved that for any tree T of order $n \geq 4$, $\gamma_{gR2}(T) \leq \gamma_{R2}(T) + 2$, and characterized all trees with $\gamma_{gR2}(T) = \gamma_{R2}(T) + 2$ and $\gamma_{gR2}(T) = \gamma_{R2}(T) + 1$.

4.14. Outer-Independent Roman $\{2\}$ -Domination

An outer-independent Roman $\{2\}$ -dominating function f on a graph G is a Roman $\{2\}$ -dominating function for which the set $\{u \in V | f(u) = 0\}$ is independent. The minimum weight of an outer-independent Roman $\{2\}$ -dominating function on G is the outer-independent Roman $\{2\}$ -domination number of G , and it is denoted by $\gamma_{oiR2}(G)$. This notion was introduced and studied in [106], and the authors called it outer-independent Italian domination. The authors provided bounds of $\gamma_{oiR2}(G)$ in terms of n , graph diameter, and vertex cover number, and they determined the exact value of $\gamma_{oiR2}(G)$ for some well-known classes of graphs. They also proved the following Nordhaus–Gaddum-type result: If G is a graph of order n , then $n - 1 \leq \gamma_{oiR2}(G) + \gamma_{oiR2}(\overline{G}) \leq 2n$. The authors provided a lower and an upper bound for $\gamma_{oiR2}(T)$, where T is a tree with $n \geq 3$; they

proved that $\frac{3n}{4} \geq \gamma_{oiR2}(T) \geq \frac{n+3+\mathcal{L}(T)}{2}$, where $\mathcal{L}(T)$ is the number of leaves in T , and they characterized trees achieving the upper bound and gave examples of trees witnessing the lower bound.

4.15. $(2, 1, 0)$ -Domination

Recall the definition of w -dominating function (see Section 4.7). Many dominating functions can be expressed as a w -dominating function for a particular vector w ; for example, if $w = (2, 0, 0)$, then the w -dominating function is precisely a Roman $\{2\}$ -dominating function. A $(2, 1, 0)$ -domination number is the minimum weight of a $(2, 1, 0)$ -dominating function. The $(2, 1, 0)$ -domination number was introduced in [107], and the authors called it $\{2\}$ -domination number, but as there is another notion in the literature with the same name, we call it $(2, 1, 0)$ -domination number and denote it by $\gamma_{(2,1,0)}(G)$ instead of $\gamma_{\{2\}}(G)$.

Note that $(2, 1, 0)$ -domination was introduced before defining Roman $\{2\}$ -domination, but as $(2, 1, 0)$ -dominating functions can be viewed as Roman $\{2\}$ -dominating functions with an extra condition, we list them in the article as a Roman $\{2\}$ -domination variable.

In [108], it was shown that the Roman $\{2\}$ -domination number of every lexicographic product $G \circ H$ can be expressed in terms of five different domination parameters of G , one of which is $\gamma_{(2,1,0)}(G)$.

In [109], the authors showed that the bound $\gamma_{(2,1,0)}(G) \geq \gamma(G) + 1$ is tight, and it was shown in [110] that $\gamma_{(2,1,0)}(G) \geq \gamma(G) + \lceil \frac{\text{diam}(G)+1}{5} \rceil$, where $\text{diam}(G)$ denotes the diameter of G . Some combinatorial results explaining the relationship between the $(2, 1, 0)$ -domination number and other domination parameters of graphs are also studied in [110]. They also proved that the $(2, 1, 0)$ -domination number of any tree T equals twice its domination number, that is, $\gamma_{(2,1,0)}(T) = 2\gamma(T)$. The authors proved that the $(2, 1, 0)$ -domination number of lexicographic product graphs $G \circ H$ depends on the value of $\gamma(H)$. For further development of w -domination or $\gamma_{(2,1,0)}$ domination, we refer to [111] and references therein.

5. Conclusions

This article is a literature review for Roman $\{2\}$ -domination (also called Italian domination), which was introduced in 2016. This notion is a variant of Roman domination, which was inspired by a strategy used to protect the Roman Empire during the reign of Emperor Constantine (272–337 AD). Since its initiation, it has received attention, and many papers have been published on Roman $\{2\}$ -domination. We surveyed all of the results on Roman $\{2\}$ -domination we found in the literature, and we also listed Roman $\{2\}$ -domination variants along with the main results of each variant. In addition, we listed open problems related to Roman $\{2\}$ -domination or its variants.

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